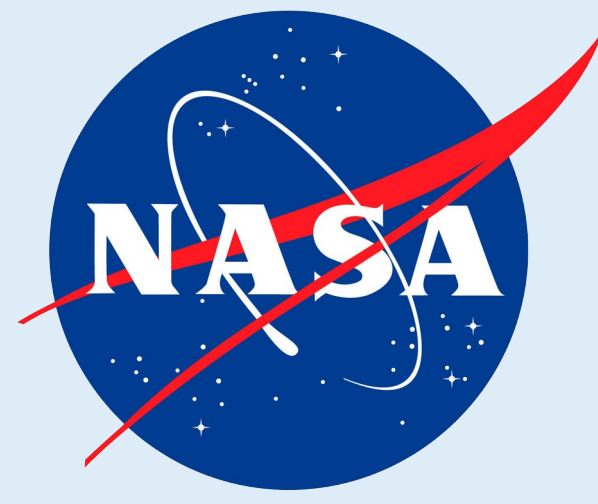




Cross-Mission expiMap Analysis Recovers Established Tissue Responses and Identifies Complementary Pathway Shifts in Mouse Spaceflight Transcriptomes

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ABSTRACT

Spaceflight affects multiple organs, but mission differences can obscure recurring biological responses. We mapped NASA OSDR mouse bulk RNA-sequencing samples into tissue-matched expiMap models trained on non-spaceflight ARCHS4 data, using about 2,000 highly variable genes linked to mouse Reactome pathways. Project-balanced shifts were evaluated with enrichment, held-out projects, three complete trainings, composition proxies, and member genes. Four tissues produced reproducible patterns: lower thymic repair and cytoskeletal programs; lower skin regulatory, repair, barrier-lipid, and cell-junction programs; lower hepatic antigen-presentation and T-cell signaling; and lower splenic T-cell, neutrophil-degranulation, and C-type lectin-receptor programs; all three splenic programs had GSEA FDR < 0.05. These results recover established responses and identify complementary tissue hypotheses to test.

INTRODUCTION

PROJECT OBJECTIVE

Identify reproducible tissue-specific gene programs across spaceflight missions.

THE CENTRAL CHALLENGE

Observed expression **Spaceflight biology** + **Tissue biology** + **Animal variation** + **Study / protocol**

Study identity and protocol can dominate an unconstrained expression representation. Tissue-matched reference mapping, accession conditioning, and equal project weighting reduce this bias without removing all confounding.

DATA AND REFERENCE MAPPING

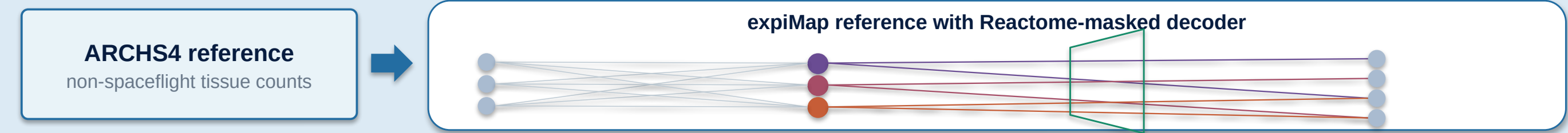
ANALYSIS SCOPE

Tissue	ARCHS4	OSDR	Projects	Programs
Thymus	1,362	117	5	387
Skin	2,593	151	4	319
Liver	5,000	197	9	364
Spleen	6,289	100	5	360

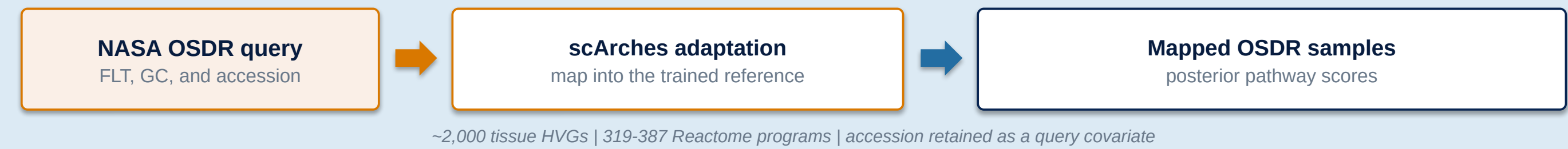
Spleen excludes nine condition-strain-confounded OSD-288 samples from the primary analysis.

METHODS: REFERENCE TRAINING AND QUERY MAPPING

1 REFERENCE TRAINING



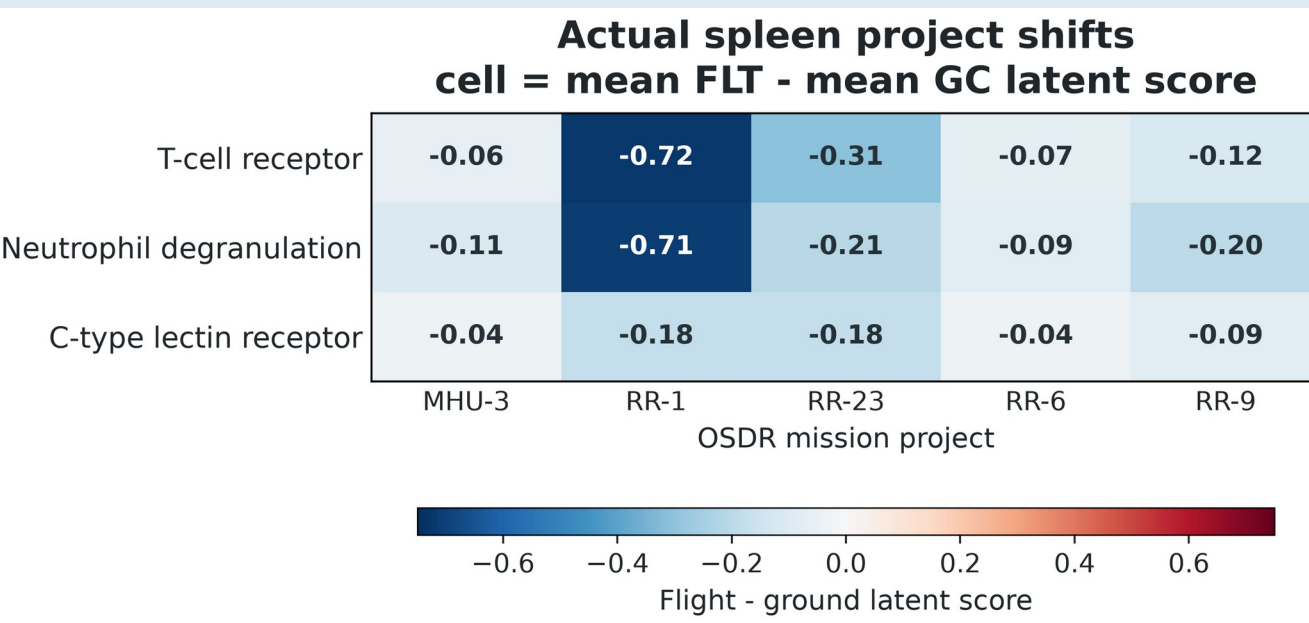
2 QUERY MAPPING



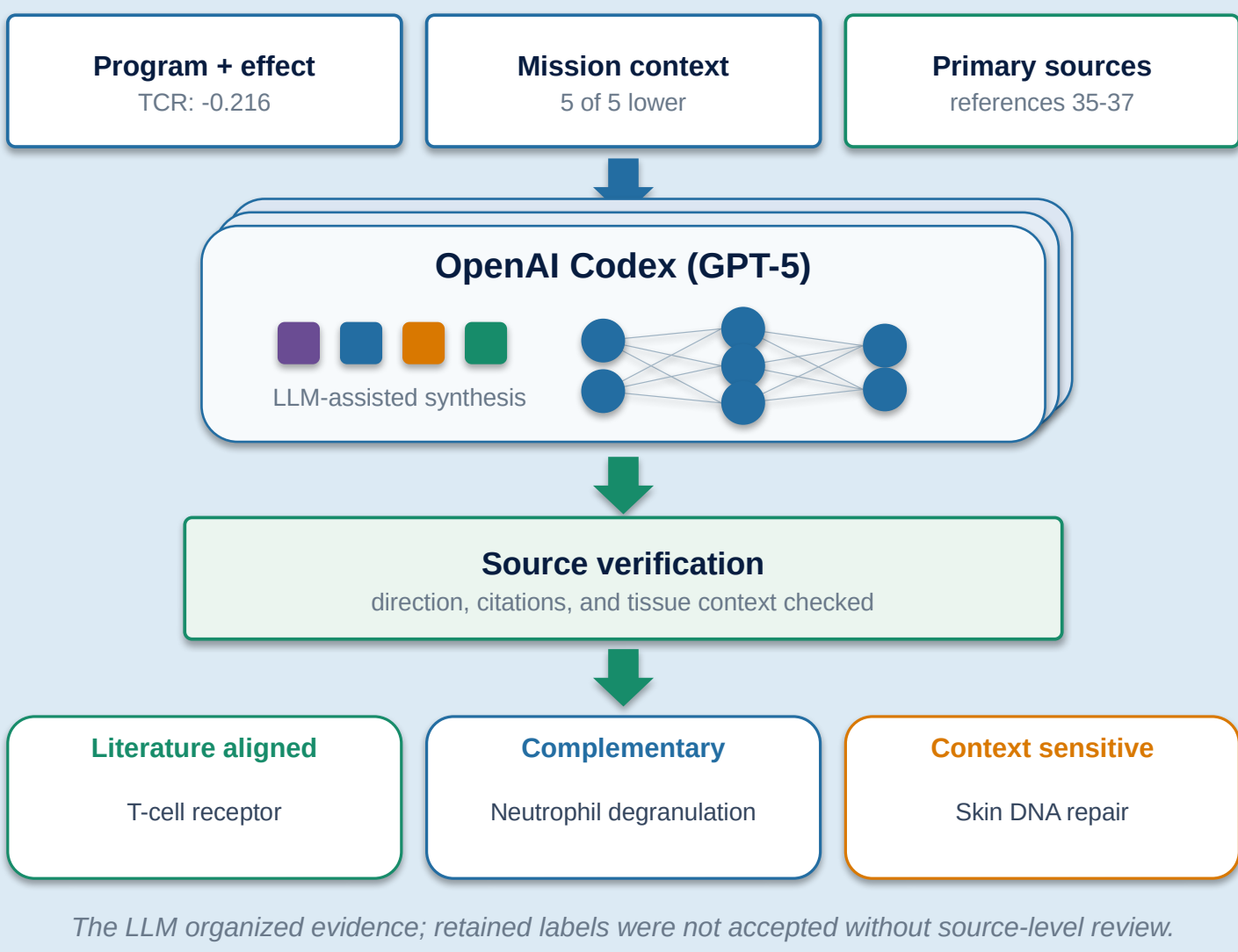
~2,000 tissue HVGs | 319-387 Reactome programs | accession retained as a query covariate

PROGRAM SCORES AND LITERATURE ANNOTATION

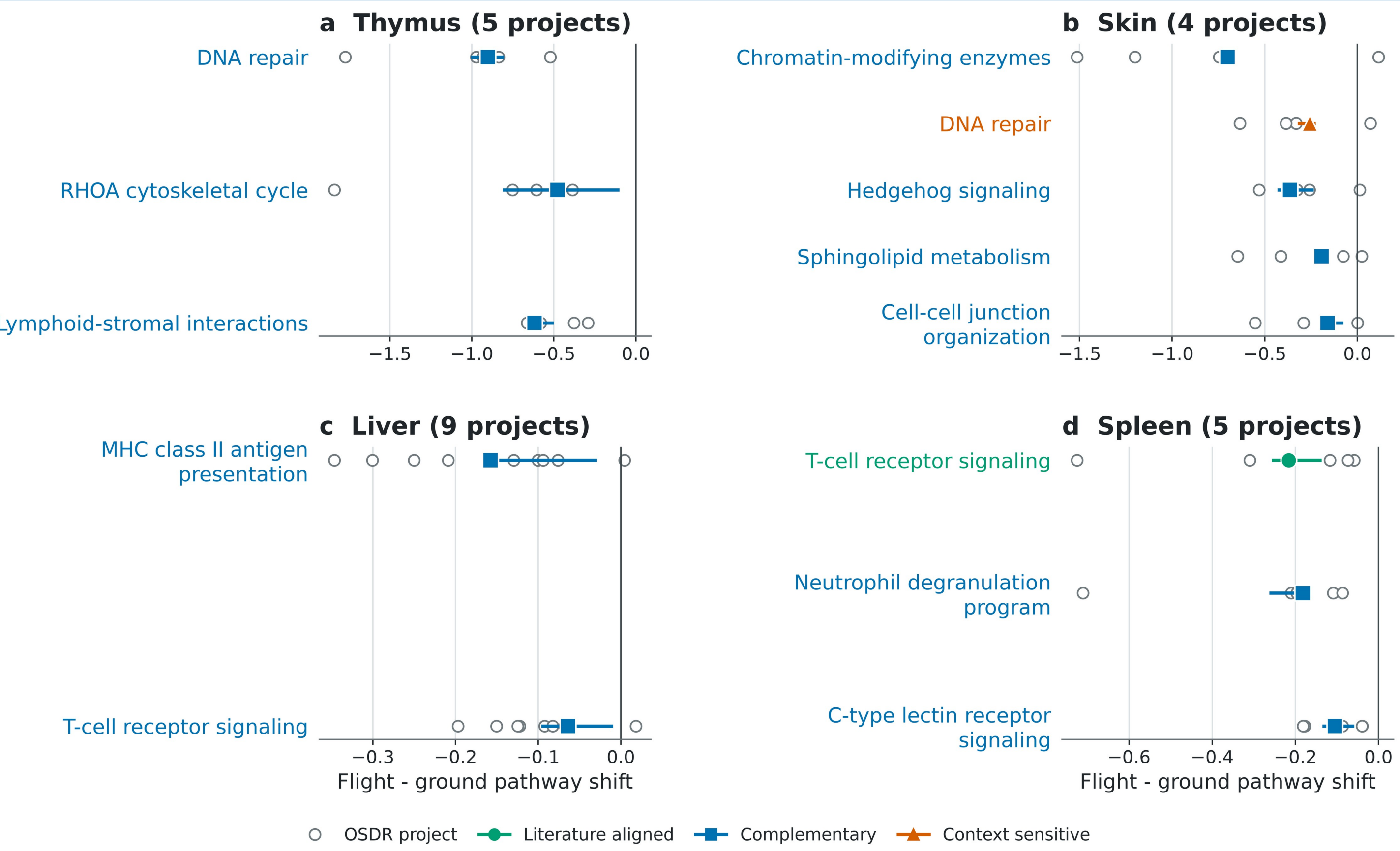
PROGRAM-SCORE SUMMARY



LLM-ASSISTED LITERATURE ANNOTATION



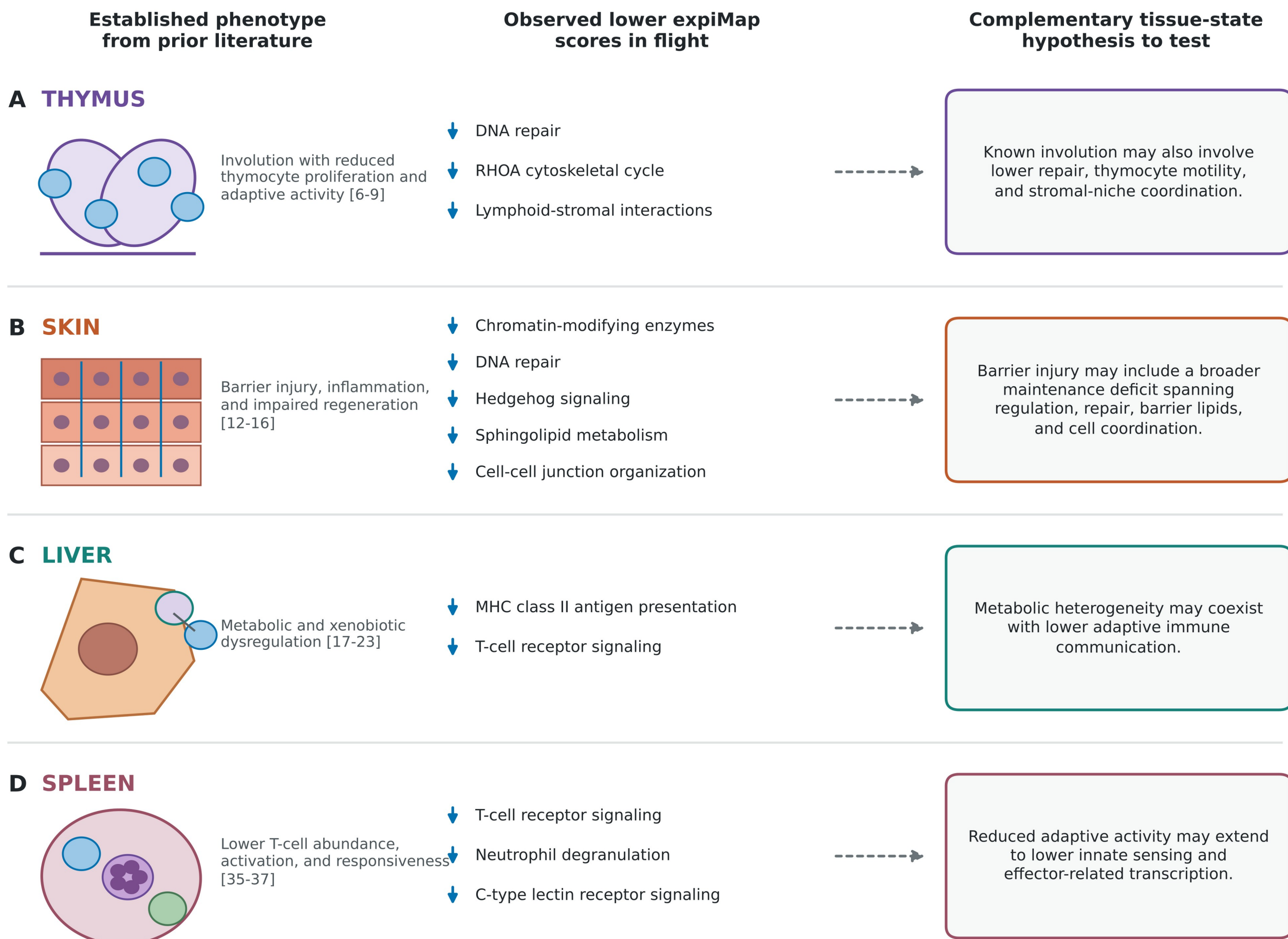
RESULTS: RETAINED PROGRAM SHIFTS



RESULTS: BIOLOGICAL INTERPRETATION

Prior evidence, observed shifts, and hypotheses to test

Prior literature, observed pathway shifts, and hypotheses to test



CONCLUSIONS

- Spleen showed the strongest evidence: three immune programs were lower across five projects and three trainings, with GSEA FDR < 0.05 for each.
- Thymus retained lower repair, cytoskeletal, and niche-interaction programs; skin retained lower regulatory, repair, regenerative, barrier-lipid, and cell-junction programs.
- Liver retained lower antigen-presentation and T-cell signaling; kidney showed a reproducible structural and growth-factor pattern but remains exploratory.
- These bulk latent shifts generate tissue-level hypotheses; they do not prove pathway inhibition or distinguish altered cell abundance from altered cell state.

REFERENCES

Selected sources; numbering follows the manuscript: [1] Gebre et al., NAR 2025; [2] Lotfollahi et al., Nat Cell Biol 2023. [4] Lachmann et al., Nat Commun 2018; [5] Milacic et al., NAR 2024; [6] Horie et al., Sci Rep 2019. [12] Cope et al., Commun Med 2024; [13] Park et al., Nat Commun 2024; [17] Beheshti et al., Sci Rep 2019. [35] Gridley et al., J Appl Physiol 2009. Complete citations and source tables accompany the manuscript.

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github.com/jasont314/nasa-mouse